EX. NO : 08

APP ARCHITECTURE – UI LAYER

AIM

To understand the UI layer architecture in Android applications and implement user interaction by separating UI components and event-handling logic.

PROCEDURE

1. Open **Android Studio** and create a new project → Empty Activity → Kotlin.
2. Design UI in **activity_main.xml** using TextView and Button.
3. Create a ViewModel class (**MainViewModel**).
4. Open **MainActivity.kt**.
5. Connect UI elements using findViewById().
6. Handle button click and fetch data from ViewModel.
7. Update TextView dynamically.
8. Run the application and observe the output.

CODE

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="20dp">
   <TextView
      android:id="@+id/tvTitle"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
```

```
        android:text="UI Layer Demo"
        android:textSize="22sp"
        android:gravity="center"/>
    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Message will appear here"
        android:textSize="18sp"
        android:paddingTop="20dp"/>
    <Button
        android:id="@+id/btnLoad"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Load Data"/>
</LinearLayout>
```

MainViewModel.kt

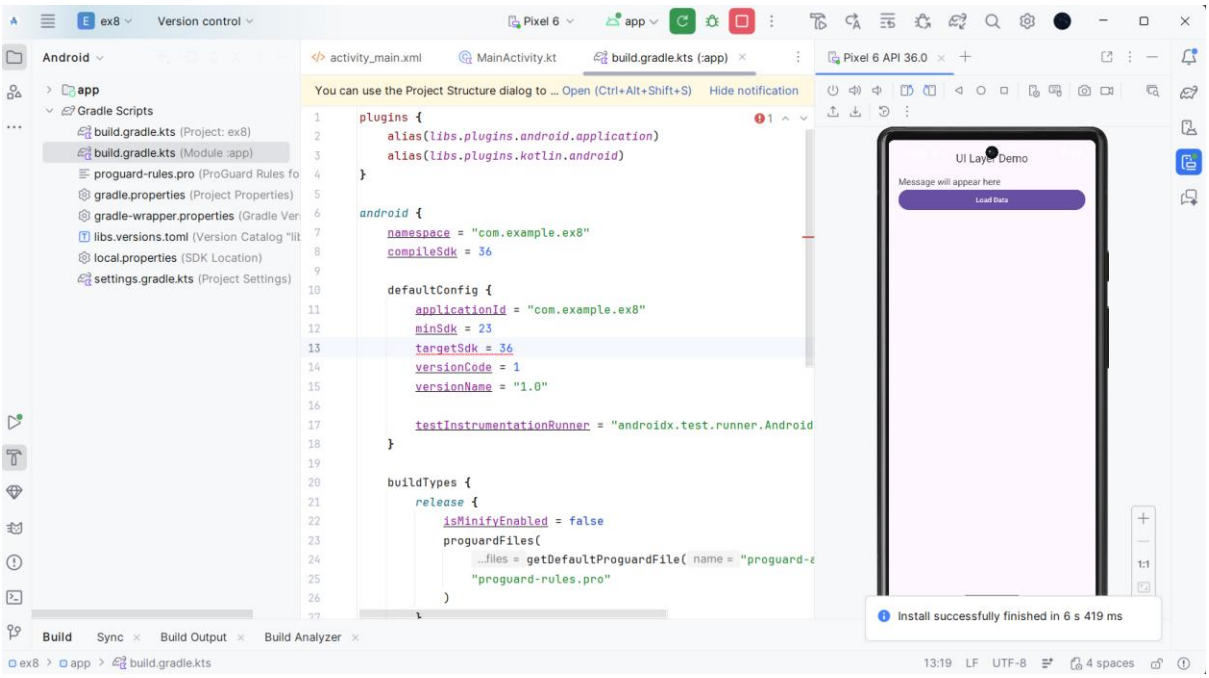
```
package com.example.uilayerdemo
import androidx.lifecycle.ViewModel
class MainViewModel : ViewModel() {
    fun getMessage(): String {
        return "Hello from ViewModel (UI Layer)"
    }
}
```

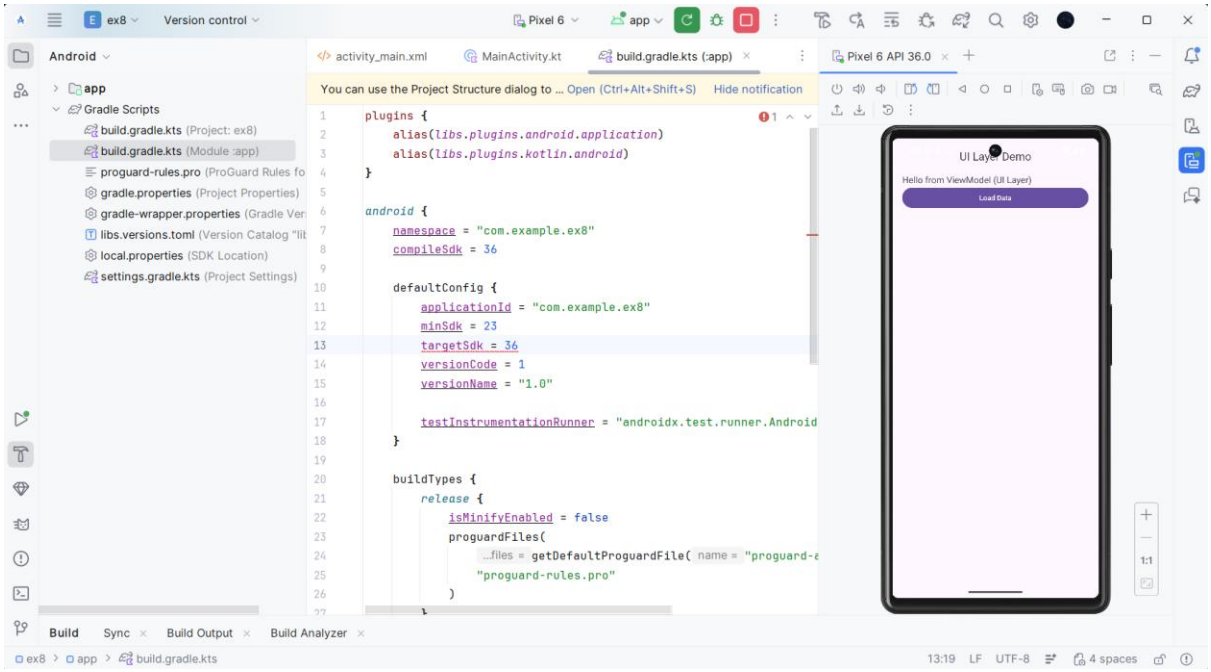
MainActivity.kt

```
package com.example.uilayerdemo
import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.activity.viewModels
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
    private val viewModel: MainViewModel by viewModels()
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        val btnLoad = findViewById<Button>(R.id.btnLoad)
        btnLoad.setOnClickListener {
            tvMessage.text = viewModel.getMessage()
        }
    }
}
```

OUTPUT





RESULT

Thus, the UI layer architecture was successfully implemented by separating UI components and logic using ViewModel, and user interaction was handled effectively in an Android application.